

CAP Theorem - Partition Tolerance

1. What is Partition Tolerance?

Partition tolerance means the distributed system continues operating even when communication between nodes breaks.

None

Network split happens

Nodes cannot fully talk

System still survives

A partition is usually caused by:

- Network failure
- Packet loss
- High latency
- Router issue
- Data center outage
- Firewall / config issue

2. Simple Meaning

Suppose 3 nodes exist:

None

Node A ↔ Node B ❌

Node B ↔ Node C ❌

Node A ↔ Node C ✅

Some nodes cannot communicate.

If system keeps functioning, it is partition tolerant.

3. Why It Matters

In real distributed systems, network failures are normal.

You cannot assume perfect communication.

So for multi-node systems:

None

Partition tolerance is practically mandatory

4. Example

Global e-commerce system:

- India region node
- Europe region node
- US region node

If Europe link fails temporarily, users in India and US should still use system.

5. CAP Tradeoff During Partition

When partition happens, system must choose:

None

Consistency OR Availability

Because partition already exists.

So CAP is really about behavior **during network partition**.

6. Option 1: CP System

Choose Consistency + Partition Tolerance

None

Reject some requests

Wait for quorum

Avoid stale data

Used for banking / metadata systems.

Examples:

- ZooKeeper
- etcd

7. Option 2: AP System

Choose Availability + Partition Tolerance

None

Continue serving requests

May return stale/conflicting data

Used for feeds / shopping catalog / analytics.

Examples:

- Apache Cassandra
- Riak

8. Why CA Is Unrealistic for Distributed Systems

If network partition occurs and system cannot tolerate it:

None

System stops

Real networks fail, so true distributed systems generally must handle partitions.

That's why many engineers say:

None

P is not optional in real distributed systems

9. How Partition Tolerance Is Achieved

A. Replication

Copies of data on multiple nodes.

B. Leader Election / Consensus

Choose healthy leader after failures.

C. Quorum Reads/Writes

Majority nodes decide.

D. Retry + Reconciliation

Sync later after network recovery.

E. Multi-Region Design

Operate despite region disconnect.

10. Example Scenario

Inventory = 5

Partition splits cluster:

None

Region A sees 5

Region B sees 5

If both sell independently:

- Available system keeps selling (risk oversell)
- Consistent system blocks one side

11. Partition Tolerance vs Fault Tolerance

Term	Meaning
Partition Tolerance	Handles network splits
Fault Tolerance	Handles node/server failures broadly

Partition tolerance is a subset of fault tolerance concerns.

12. HLD Design Use Cases

Must consider partition tolerance in:

None

Microservices

Multi-region apps

Distributed DBs

Kafka clusters

Kubernetes control planes

Payment gateways

13. Interview Answer Example

Q: What is partition tolerance in CAP theorem?

Partition tolerance means the system continues operating despite communication failures or message loss between nodes in the distributed network.

14. Short Revision Notes

None

Partition Tolerance:

system survives network split

During partition choose:

CP or AP

Real distributed systems need P

15. One-Line Summary

Partition tolerance means a distributed system continues functioning despite network communication failures between nodes, forcing tradeoffs between consistency and availability.